SUVIDH MATHUR

Bachelor of Technology
in Artificial Intelligence and Data Science
MBM University, Jodhpur

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Education

Year	Degree/Certificate	Institute	CGPA/Percentage
2023-2026	B.Tech Artificial Intelligence and Data Science (Lateral Admission)	MBM University, Jodhpur	78%
2020-2023	Diploma in Civil Engineering	Government Polytechnic College, Jodhpur	84%

Experience

- **AI/ML Developer | MineExcellence, Jodhpur** Jan 2026 - June 2026
  - Developed algorithms for structural feature detection and geological data analysis from 3D Point Cloud data, processing 2M+ points per scan.
  - Processed large-scale point cloud data and generated 10+ structural metrics and visualizations for each detected discontinuity, including stereonet projections, density contours, crack heatmaps, and automated engineering reports.
- **Data Analyst & Application Developer | Secure Meters Limited, Udaipur** Jun 2025 - Jul 2025
  - Designed and developed CUBE, a Service & Repair (SR) Management application, enabling data-driven insights through structured analytics and visualization.
  - Implemented data cleaning, preprocessing, and transformation pipelines using Python to support business intelligence and decision-making through interactive dashboards.
- **Research Intern, Deep Learning & Computer Vision | DRDO, Jodhpur** May 2024 - Aug 2024
  - Worked on Deep Neural Networks, RCNN, and YOLO-based object detection models, applying computer vision techniques to real-world defense use cases.
  - Designed and developed a Trash Detection system using YOLOv8, including dataset preparation, annotation, training, and performance evaluation.

Projects

- **CUBE – Service & Repair (SR) Management & Analytics Application**
  Python, ETL, Plotly, Data Analytics **GitHub**
  - Designed and developed CUBE, an end-to-end Service & Repair Management and Analytics application that automated the generation of 4 daily service reports, reducing processing time by up to 80%.
  - Automated ETL pipelines to transform raw service data into analysis-ready datasets.
  - Built interactive Plotly dashboards to monitor 10+ operational KPIs and deliver real-time insights.
- **DR.MIGI – AI Healthcare Assistant**
  LLMs, RAG, Vector Embeddings, Semantic Search **GitHub**
  - Developing DR.MIGI, an AI-powered Healthcare Assistant using Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) for context-aware medical assistance.
  - Built an intelligent RAG pipeline integrating Vector Embeddings, Semantic Search, and Context-Aware question answering for medical report analysis.
  - Designed a modular backend architecture for LLM inference, document retrieval, and future EHR and medical device integration.
- **Trash Detection Systems using YOLOv8**
  YOLOv8, Python, OpenCV, Computer Vision **GitHub**
  - Designed a real-time trash detection system using YOLOv8, Python, and OpenCV for object detection in real-world environments.
  - Performed dataset collection, annotation, preprocessing, training, validation, and model evaluation.
  - Improved detection performance through hyperparameter tuning and data augmentation, achieving an average detection confidence of 0.80.

Technical Skills

- **Programming Languages:** Python, SQL, Object-Oriented Programming (OOP)
- **AI/ML:** Machine Learning, Deep Learning, NLP, Computer Vision, YOLOv8, RCNN, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Faster-Whisper
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, Plotly, FastAPI
- **Software Development:** REST APIs, API Development, Backend Development, Git, GitHub, Version Control, Database / SQL
- **Data & Analytics:** Data Analysis, Data Cleaning, Data Preprocessing, ETL Pipelines, Data Visualization, Power BI
- **Tools & Platforms:** Roboflow, Potree, Arduino

Achievements & Certifications

- **1st Place – EDIFICIO'24, IIT Jodhpur:** Built an IoT-enabled Hyperloop prototype using ESP32 microcontrollers and wireless control.
- AI/ML Assessment – Silicon Interfaces (B2Cie™), 2026
- SQL (Basic) – HackerRank, 2026
- Power BI – Microsoft & TCS (Forage), Data Visualization & Business Intelligence
- Deloitte (Forage) – Data Visualization with Tableau
- Google Cloud Platform Fundamentals – Simplilearn